MILITARY SPECIFICATION

WINTERIZING KIT, PROTECTIVE MASK, M1

1. SCOPE

1.1 Scope. This specification covers a protective mask winterizing kit (see 6.1).

1.2 Classification. Kits shall be of the following styles, as specified (see 6.2):

Style 1 - Left side, for use with protective masks having the canister mounted on the wearer's left.
Style 2 - Right side, for use with protective masks having the canister mounted on the wearer's right.

2. APPLICABLE DOCUMENTS

2.1 Government documents. The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of the specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

QQ-S-781 - Steel, Strapping, Flat.
TT-W-572 - Wood Preservative; Water-Repellent.
PPP-B-566 - Boxes, Folding, Paperboard.
PPP-B-621 - Boxes, Wood, Nailed and Lock-Corner.
PPP-B-636 - Box, Fiberboard.
PPP-F-320 - Fiberboard, Corrugated and Solid, Sheet Stock (Container Grade), and Cut Shapes.

MILITARY

MIL-B-2427 - Boxes, Ammunition Packing; Wood Nailed.

STANDARDS

FEDERAL

FED-STD-406 - Plastics: Methods of Testing.

Source: https://assist.dla.mil -- Downloaded: 202640-22T20:57Z
Check the source to verify that this is the current version before use.

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.

DRAWINGS

US ARMY MUNITIONS COMMAND

EDGEWOOD ARSENAL

DL5-74-66 - Winterizing Kit, Protective Mask, M1.
D6540-A50-0089 - Lensometer, Q44.

PUBLICATIONS

EDGEWOOD ARSENAL INSTRUCTION MANUAL

136-300-118 - Simple, Ophthalmic Lens; Operation and Maintenance of Shadowgraph.

(Copies of specifications, standards, drawings, and publications required by the suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Materials and components.

3.1.1 Materials. All materials cited on Drawing DL5-74-66 or on the subsidiary drawings, shall conform to the specifications listed thereon, or to the specific characteristics set forth on the drawings.

3.1.2 Components. All components of the winterizing kit shall conform to the specifications and drawings listed on Drawing DL5-74-66 and subsidiary drawings.

3.2 Assembly. The winterizing kit shall be assembled as specified on Drawing D5-74-66.

3.3 Cements for cheek pad. The following cements shall be used to assemble the cheek pads (see Drawing C5-74-31).

3.3.1 Cement (flannel-to-butyl coated cloth). The following formula for waterproof, low temperature, rubber cement is used to secure the

Source: https://assist.dla.mil -- Downloaded: 202610-22T20:57Z
Check the source to verify that this is the current version before use.

canton flannel insert to the butyl coated cloth cheek pad blank and to secure the two cheek pad blanks together:

Cement	Percent (+ 0.1) by weight
*Dry Formula	24.1
n-Hexane	72.3
Dithiocarbamate	3.6

*Dry Formula	Percent (+0.01) by weight
Smoked Sheets	72.60
Calcium Carbonate, Precipitated	21.55
Zinc Oxide	3.63
Carbon Black, Natural Gas (Medium thermal black)	.22
Symmetrical Dibeta-naphthyl-para-phenyl-diamine	.73
Sulfur	1.27

3.3.2 Cement (cheek pad blank-to-backing sheet). The following formula for rubber cement is used to secure the butyl coated cloth cheek pad to the holland cloth backing sheet:

Cement	Percent (+ 0.1) by weight
*Dry Formula	19.0
Naphtha (Boiling Range 80° to 100°C)	80.0
Ethyl Alcohol	1.0

*Dry Formula	Percent (+ 0.01) by weight
Smoked Sheets	71.43
Coumarone-Indene Resin (Melting Point 20° to 30°C)	25.00
Zinc Oxide	3.57

3.4 Lens (Drawing C5-74-82).

3.4.1 Prismatic effect. The prismatic effect of the lens shall not exceed one-eighth diopter in the vertical meridian nor more than three-eighths diopter in the horizontal meridian when tested as specified in 4.4.4.1.

3.4.2 Refractive power. The refractive power of the lens shall not exceed plus or minus one-eighth diopter when tested as specified in 4.4.4.2.

3.4.3 Local distortion. The lens shall exhibit no regions wherein the difference in prismatic effect or refractive power exceeds one-eighth

Source: https://assist.dla.mil -- Downloaded: 202640-22T20:57Z
Check the source to verify that this is the current version before use.

diopter in any meridian when tested as specified in 4.4.4.3.

3.4.4 Local defects. When tested as specified in 4.4.4.4, any local defects present in the lens shall satisfy the following requirements:

3.4.4.1 Number. The total number of detectable local defects in the lens shall not exceed eight.

3.4.4.2 Proximity. There shall be no two detectable local defects separated by less than 1/4 inch.

3.4.5 Light transmission. The lens shall transmit a minimum of 89 percent of the incident visible light when tested as specified in 4.4.4.5.

3.4.6 Haze. Haze shall be no more than 4.0 percent when tested as specified in 4.4.4.5.

3.5 Preproduction. Prior to the start of regular production, a preproduction sample of winterizing kits shall be produced in accordance with this specification for examination and tests (see 4.3).

3.6 Workmanship.

3.6.1 Lens (clear). The lens shall be colorless, smooth, and free from blemishes, striae, waves, and damage such as cracks, splits and other visible defects as revealed in shadowgraph examination (see 4.4.4.4) which will impair its appearance or cause distortion of vision. In addition the lens shall be free from foreign matter. Workmanship and local defects of a magnitude greater than the referee standards established by the Government (see 6.4) for Major and Minor defects shall be categorized under the appropriate defect class.

3.6.2 Winterizing kit. The kit shall be free from damage such as cuts, tears, punctures, loose or missing stitches, and foreign matter such as dirt, grease, or oil.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.

4.1.1 Supplier's responsibility. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to specified requirements.

Source: https://assist.dla.mil -- Downloaded: 202640-22T20:57Z
Check the source to verify that this is the currentversion before use.

4.1.2 Objective evidence. The supplier shall provide objective evidence acceptable to the contracting officer that the requirements of 3.1, 3.3, and section 5 for which specific inspection has not been provided in this specification have been satisfied.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

(a) Preproduction inspection (see 4.3).
(b) Quality conformance inspection (see 4.4).

4.3 Preproduction inspection.

4.3.1 Sample. A preproduction sample of ten winterizing kits shall be manufactured using the same methods, materials, equipment and processes as will be used during regular production.

4.3.2 Inspection procedures.

4.3.2.1 For examination. The preproduction sample of winterizing kits shall be examined in accordance with the classification of defects 4.4.3.3. One set of lens (Drawing C5-74-82) from each kit shall be used for the examination of the lens.

4.3.2.2 For test. One set of lens (Drawing C5-74-82) taken at random from each preproduction sample of winterizing kits shall be tested in accordance with 4.4.4.

4.3.3 Acceptance/rejection criteria. The preproduction sample shall comply with the examinations and tests specified in 4.3.2 to be acceptable. The supplier shall obtain written approval from the contracting officer prior to proceeding with regular production.

4.4 Quality conformance inspection.

4.4.1 Lotting. A lot shall consist of the winterizing kits produced by one manufacturer, at one plant, from the same materials, under essentially the same manufacturing conditions, and submitted for acceptance at one time.

4.4.2 Sampling.

4.4.2.1 For examination and nondestructive tests. Sampling shall be conducted in accordance with MIL-STD-105.

4.4.2.2 For destructive test. Sampling shall be conducted in accordance with MIL-STD-105, level S-3.

4.4.3 Inspection procedures.

Source: https://assist.dla.mil -- Downloaded: 202610-22T20:57Z
Check the source to verify that this is the current version before use.

4.4.3.1 For examination and nondestructive tests. Sample winterizing kits and the level A preparation for delivery shall be examined and tested in accordance with the classification of defects 4.4.3.3 and MIL-STD-105. One set of lens (Drawing C5-74-82) selected at random from each sample kit shall be used for the specified examination and testing of the lens.

4.4.3.2 For destructive test. One set of lens (Drawing C5-74-82) selected at random from each sample winterizing kit shall be tested in accordance with 4.4.4.5 and MIL-STD-105, level S-3 using an AQL of 0.65 percent defective for acceptance.

4.4.3.3 Classification of defects.

(a) Winterizing kit, protective mask, M1 (Draing D5-74-66).

Categories		Defects
Critical:		None defined
Major:	AQL	2.5 percent defective
101		Component missing
*102		Components not conforming to pattern as delineated on applicable drawings
103		Hardware incorrect or nonfunctional
104		Cheek pad assembled incorrectly
105		Stitching not as specified
106		Workmanship (see 3.6.2)
Minor:	AQL	4.0 percent defective
201		Marking missing, incorrect, or illegible
202		Seams turned incorrectly

*Inspection to be performed prior to assembly.

(b) Lens (Drawing C5-74-82).

Categories		Defects	Acceptance standards
Critical:		None defined	
Major:	AQL	0.65 percent defective	
101		Prismatic effect excessive	4.4.4.1
102		Refractive power out of limits	4.4.4.2
103		Distortion	4.4.4.3
104		Local defects	4.4.4.4

Source: https://assist.dla.mil -- Downloaded: 202640-22T20:57Z
Check the source to verify that this is the currentversion before use.

4.4.3.3 Classification of defects (continued).

(b) Lens (Drawing C5-74-82).

Categories	Defects	Acceptance standards
	AQL 1.0 percent defective	
105	Hole location not as specified (right hand or left hand)	
106	Thickness	
107	Overall dimensions	
108	Radius of curvature	
109	Workmanship (see 3.6.1)	4.4.4.4
110	Radii incorrect	
Minor:	AQL 2.5 percent defective	
201	Workmanship (see 3.6.1)	4.4.4.4

(c) Preparation for delivery (Section 5).

Categories	Defects
Critical:	None defined
Major:	AQL 4.0 percent defective
101	Packaging box not as specified
102	Closure of packaging box incorrect
103	Packaging box damaged
104	Outer container not as specified
105	Internal cushioning missing or not as specified
106	Incorrect quantity of unit containers in exterior boxes
107	Boxes not coated with wood preservative, when applicable
108	Marking missing, incorrect, or illegible

4.4.4 Tests.

4.4.4.1 Prismatic effect. The prismatic effect of the lens shall be determined using the Q44 Lensometer (D6540-A50-0089).

4.4.4.2 Refractive power. The refractive power of the lens shall be determined using the Q44 Lensometer (D6540-A50-0089).

Source: https://assist.dla.mil -- Downloaded: 202640-22T20:57Z
Check the source to verify that this is the current version before use.

4.4.4.3 Local distortion. Any region, which in the course of visual inspection or examination shows evidence of distortion due to irregularity of either surface, inhomogeneity of plastic materials, surface or internal defects, shall be examined in the following manner: Measure the apparent prismatic effect and refractive power (4.4.4.1 and 4.4.4.2) of the affected region, and then the prismatic effect and refractive power of the region adjacent to the affected region, and check for compliance with 3.4.3.

4.4.4.4 Local defects. The lens shall be examined for compliance with 3.4.4 by comparative shadowgraph in accordance with Instruction Manual 136-300-118.

4.4.4.5 Light transmission and haze. The light transmission and haze shall be determined using Method 3022 of FED-STD-406.

5. PREPARATION FOR DELIVERY

5.1 Packaging, levels A, B and C. Each kit (Drawing D5-74-66) shall be packaged in a paperboard box conforming to variety 2, style II modified tuck end, type A, class a or b of PPP-B-566. The inside dimensions of the box shall be 5 by 1-3/4 by 7 inches length, width, and depth respectively. Closure shall be in accordance with the appendix to PPP-B-566.

5.2 Packing. Level shall be A, B, or C as specified (see 6.2).

5.2.1 Level A. Seventy-two kits, packaged as specified in 5.1, shall be packed in a nailed wood box conforming to class 2 (overseas), style 4, type 2 load of PPP-B-621. The box shall have inside dimensions of 34 by 14-3/8 by 10-3/8 inches, length, width and depth respectively. The kits shall be arranged in two layers of two rows of 18 each. The layers shall be separated with a fiberboard pad conforming to type CF, variety SW, grade W5c or W6c of PPP-F-320, having approximate dimensions of 33-1/2 by 14-1/8 inches. Filler pads, conforming to the same material as the separator pads, shall be used either in the sides, ends, top or bottom to restrict free movement of the packaged kits. The box shall be closed and strapped in accordance with the closure requirements for class 2 boxes in the appendix to PPP-B-621. Strapping shall conform to type I, class B of QQ-S-781. The boxes or the finished wood parts thereof shall be preserved by completely immersing for a minimum of three minutes in a water repellent wood preservative conforming to composition A of TT-W-572 and then dried. Prior to packing items in the box, the treated wooden box shall meet the dryness of preservative requirement specified in MIL-B-2427.

5.2.2 Level B. Seventy-two kits, packaged as specified in 5.1, shall be packed in accordance with 5.2.1 except that the boxes shall not be preservative treated and strapping shall conform to type I, class A of QQ-S-781.

Source: https://assist.dla.mil -- Downloaded: 202640-22T20:57Z
Check the source to verify that this is the current version before use.

5.2.3 Level C. Seventy-two kits, packaged as specified in 5.1, shall be packed in accordance with 5.2.1 except that the box shall be fiberboard conforming to type CF, style RSC, class domestic, variety SW, grade 200 of PPP-B-636. Separator and filler pads shall be type CF, variety SW, grade 125 of PPP-F-320. Closure shall be in accordance with the appendix to PPP-B-636.

5.3 Marking. The unit packages and shipping containers shall be marked "Left Side" for style 1 kits and "Right Side" for style 2 kits. In addition to any special marking required by the contract or order, unit packages and shipping containers shall be marked in accordance with MIL-STD-129. Sufficient drying time shall be allowed after the application of the wood preservative in 5.2.1 to insure that all markings shall be clear, legible, and permanent.

6. NOTES

6.1 Intended use. This specification covers a winterizing kit intended for use with the M9 type masks at low temperatures down to -40°F.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Style of kit required (see 1.2).
(c) Type of packaging and packing required (see 5.1 and 5.2).
(d) Preproduction:

(1) Time allowed for supplier submission of samples for Government test and evaluation after award of contract.
(2) Name and address of test facility and shipping instructions when testing is performed by the Government.
(3) Time required for the Government to notify the supplier whether or not to proceed with production.

6.3 Supersession data. This specification includes the requirements of Chemical Corps Purchase Description 197-54-638, dated 12 May 1958.

6.4 Referee standards. The Government will furnish referee standards to define the type and category (Major and Minor) of each defect. These referee standards are based upon the defect magnitude and the potential deleterious effect on the wearer.

Custodian:

Army - MU

Preparing activity:

Army - MU(EA)

Project No. 4240-A426

Source: https://assist.dla.mil -- Downloaded: 202640-22T20:57Z
Check the source to verify that this is the currentversion before use.